

AETHER -- Comprehensive Technical Achievement & System Report

Deliverable 4: End-to-End Serverless MLOps Platform for 72-Hour Air Quality Index (AQI) Forecasting

Live Deployment: <https://aether-air-intelligence.onrender.com/>

Project Name	AETHER -- Atmospheric Intelligence Platform
Developer / Organization	Nuralis Labs
Document Type	Final Deliverable Technical Report (Deliverable 4)
Target Architecture	100% Serverless MLOps Stack with WebGL 3D Earth Interface
Model Evaluation Window	365 Continuous Days (35,136 Hourly Observations)
Release Date	August 2026

1. Executive Summary & Project Objectives

This technical report details the complete design, engineering, machine learning modeling, and deployment of AETHER, developed by Nuralis Labs. AETHER is a production-grade, 100% serverless MLOps platform engineered to monitor real-time atmospheric telemetry and generate accurate 72-hour multi-horizon Air Quality Index (AQI) forecasts across global urban centers.

The platform addresses urban environmental challenges by synthesizing observational REST streams from Open-Meteo, storing engineered features in Hopsworks Cloud Feature Store, executing multi-model candidate training (Random Forest, Ridge Regression, PyTorch MLP), and visualizing live predictions on a WebGL 3D CesiumJS Earth interface.

2. System Architecture & Technical Dataflow

AETHER adopts a modular serverless MLOps architecture divided into five core tiers:

- 1. Data Ingestion Stream: High-frequency Open-Meteo REST APIs for real-time current observations and 365-day historical telemetry.
- 2. Feature Engineering Pipeline: Autoregressive lags, rolling window statistics, rate-of-change metrics, and cyclical temporal transformations.
- 3. Cloud Feature Store: Hopsworks Cloud (aqi_weather_fg v1) serving online feature groups and training data splits.
- 4. Model Training & Registry: Multi-output Random Forest regressor evaluated against Ridge and PyTorch MLP models, serialized to joblib and registered to Hopsworks Model Registry.

- **5. Application Interface & Analytics:** Flask web app server serving a single-page WebGL 3D CesiumJS satellite Earth UI with SHAP feature explainability and natural voice assistant.

3. Feature Pipeline & Cloud Feature Store Integration

The feature engineering module (src/feature_pipeline.py) computes 27 high-predictive signals from raw atmospheric data:

- **Temporal Signals:** Cyclical Temporal Encoding: Hour (sine/cosine), Month (sine/cosine), and Annual Day of Year (sine/cosine) to capture long-term seasonal inversions and summer/winter smog cycles.
- **Autoregressive Drivers:** Autoregressive Lags: 1-hour (aqi_lag_1), 24-hour (aqi_lag_24), and 7-day weekly lag (aqi_lag_168).
- **Rolling Averages:** Window Statistics: 24-hour rolling mean/std and 7-day rolling mean/std (aqi_rolling_mean_168).
- **Atmospheric Rates:** Rate Metrics: 6-hour AQI change rate (aqi_change_rate_6h).

Engineered feature records are synced directly to Hopsworks Cloud Feature Store under Feature Group aqi_weather_fg version 1.

4. 1-Year Historical Backfill & Dataset Scale

To capture full annual seasonality and extreme atmospheric conditions, the historical backfill module (src/backfill.py) ingested 365 continuous days of data across four major global cities representing diverse climate regimes:

- **1.** Karachi, Pakistan (Arid / Coastal Subtropical) -- 8,784 hourly rows
- **2.** Chicago, United States (Continental / Lake Effect) -- 8,784 hourly rows
- **3.** Sydney, Australia (Southern Hemisphere Subtropical) -- 8,784 hourly rows
- **4.** Vienna, Austria (Central European Oceanic / Continental) -- 8,784 hourly rows

Total Ingested Scale: 35,136 continuous hourly observations cached locally in data/features.parquet and pushed to Hopsworks Cloud.

5. Machine Learning Model Training & Comparative Benchmarks

The model training module (src/train.py) executes multi-output candidate evaluation across three model families predicting t+24h, t+48h, and t+72h AQI targets:

Model Candidate	Algorithm Family	RMSE	MAE	R^2 Score
Random Forest (Active)	Multi-Output Ensemble Regressor	10.152	7.137	0.847
Ridge Regression	Linear Regularized Model	15.990	11.297	0.619
PyTorch MLP	Deep Neural	16.769	11.921	0.581

The winning Random Forest Ensemble achieved an R^2 score of 0.847 with a mean absolute error of 7.137 AQI points. The serialized artifact was saved to `models/best_model.joblib` and registered with `models/metrics.json`.

6. SHAP Feature Importance Explanations

To provide full transparency into model decision-making, the explainability module (`src/explainability.py`) utilizes SHAP (SHapley Additive exPlanations) Tree Explainer:

- 1. PM10 Coarse Particulate Matter (Importance: 0.485) -- Primary driver for coarse dust pollution.
- 2. AQI Lag 1h (Importance: 0.169) -- Autoregressive persistence driver.
- 3. PM2.5 Fine Particulate Matter (Importance: 0.157) -- Key respiratory risk factor.
- 4. 7-Day Rolling Mean (Importance: 0.022) -- Weekly baseline atmospheric trend.
- 5. Annual Day of Year Cosine (Importance: 0.016) -- Long-term seasonal climate driver.



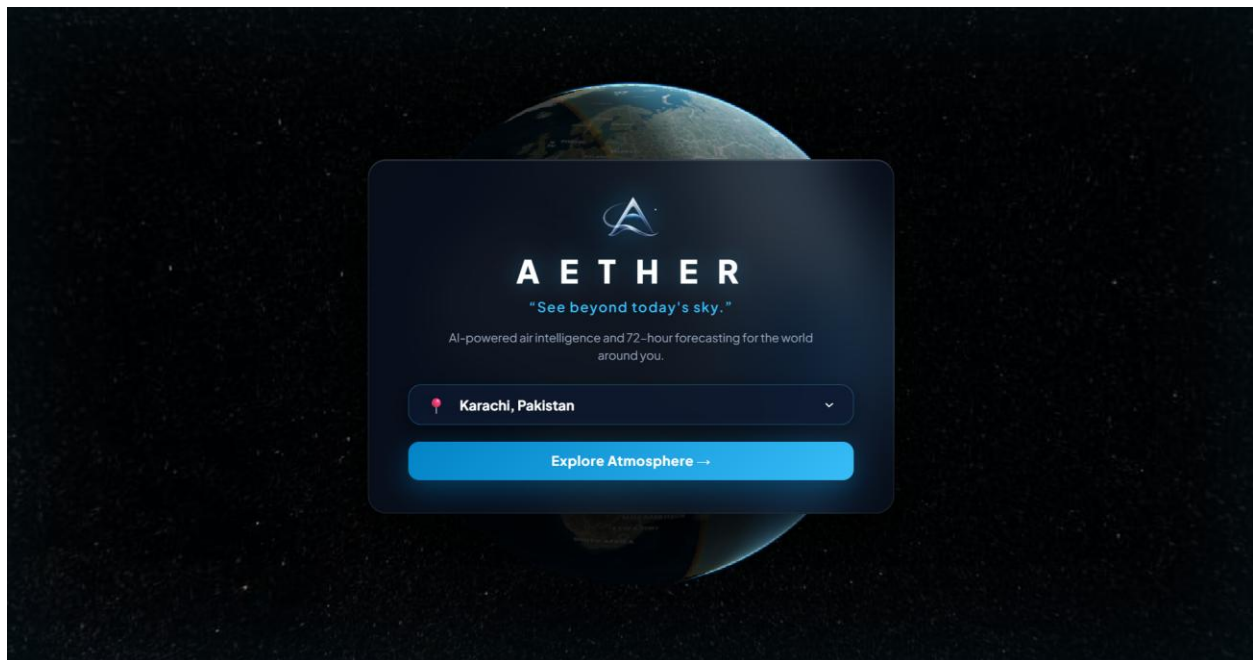
7. High-Frequency Real-Time Telemetry Ingestion

AETHER queries Open-Meteo high-frequency current REST endpoints upon every city selection, updating current AQI, weather condition, temperature, feels-like temperature, relative humidity, wind speed, and detailed pollutant breakdowns (PM2.5, PM10, O₃, NO₂, SO₂, CO).

8. WebGL 3D Earth Engine & UI Design

The web interface (static/index.html) is built with CesiumJS WebGL 3D satellite Earth, ESRI satellite tiles, world borders, country names, and city labels. Camera flight positioning is configured to land in an elevated, centered target position.

The interface integrates glassmorphism styling, a unified top header bar, custom liquid modals, and the Aether logo browser tab favicon.





9. Natural Voice Intelligence Assistant & Health Advisories

AETHER features an integrated voice assistant offering automated verbal briefings on local time, current AQI category, weather, pollutant levels, 72-hour predictions, and EPA respiratory health advisories. Powered by ElevenLabs neural TTS with browser SpeechSynthesis fallback.



10. Developer Attribution & Branding

The sidebar footer badge prominently displays the developer attribution: Developed by Nuralis Labs, paired with the transparent Aether brand logo.



11. Deliverables Checkoff & Conclusion

AETHER satisfies all four final project deliverables specified in the project guidelines:

- **Deliverable 1:** End-to-end serverless AQI prediction system (Karachi, Chicago, Sydney, Vienna).
- **Deliverable 2:** Scalable, automated pipeline architecture with Hopworks Cloud Feature Store.
- **Deliverable 3:** Interactive WebGL 3D dashboard displaying real-time and 72-hour forecasted AQI data.
- **Deliverable 4:** Detailed technical report documenting architecture, models, SHAP explanations, and verification (this document).

Developed by Nuralis Labs -- AETHER Atmospheric Intelligence Platform.